

Universidad Distrital Francisco José de Caldas

Computer Engineering Program — School of Engineering

Systems Analysis & Design 2026-I

Academic Collaboration Network: A Systems Engineering Approach to Mitigate Isolated Learning Through Skill-Based Group Optimization

Technical Report — Workshops 1, 2 & 3

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Date: June 2026

Version: 3.0 — Final

Repository: <https://github.com/AnyeloCZ/Academic-Collaboration-Network-SAD-2026-I>

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Glossary and Acronyms

ACN	Academic Collaboration Network
API	Application Programming Interface
CPM	Critical Path Method
GDPR	General Data Protection Regulation
GBI	Group Balance Index
HPA	Horizontal Pod Autoscaler
JWT	JSON Web Token
LMS	Learning Management System
MFA	Multi-Factor Authentication
MVP	Minimum Viable Product
NPS	Net Promoter Score
PDCA	Plan-Do-Check-Act
PMBOK	Project Management Body of Knowledge
RACI	Responsible, Accountable, Consulted, Informed
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SIA	Sistema de Información Académica (university grading system)
SSO	Single Sign-On
SUS	System Usability Scale
TLS	Transport Layer Security

Abstract

Higher education systems in Colombia and globally face persistent structural inefficiencies in peer-learning organisation: spontaneous, socially driven group formation produces skill clustering, isolated learners, and inequitable knowledge distribution. This report documents the full research and engineering lifecycle of the **Academic Collaboration Network (ACN)**, a socio-technical platform designed to formalise study group formation as a constrained optimisation problem supported by graph modelling and algorithmic decision mechanisms. Primary data collected via structured survey ($n = 25$ university students at Universidad Distrital Francisco José de Caldas, March 2026) confirmed that 88% rely exclusively on WhatsApp for academic coordination, 68% cite communication or interest deficits as the main barrier, and 96% expressed positive or conditional adoption intent ($\bar{x} = 3.88/5$). Building on this evidence, Workshop 2 produced a microservices architecture with seven core services targeting < 2 s response times, 99.5% availability, and support for 500+ concurrent users. Workshop 3 extended the design to a production-ready plan incorporating ISO 9001 and IEEE 730 quality frameworks, a ten-risk PMBOK/ISO 31000 register, full critical path analysis, and a \$50,000 USD seven-month delivery roadmap. Experimental simulation over 200 synthetic profiles confirmed that complementary variance minimisation significantly reduces intra-group skill dispersion while increasing network density and centralityequity. This report consolidates all three workshops into a single cohesive technical record, demonstrating clear evolution from exploratory analysis through structural design to operational implementation readiness.

1 Introduction

Higher education institutions increasingly promote collaborative learning as a mechanism to improve academic performance, critical thinking, and knowledge retention [1]. Despite broad institutional endorsement of teamwork, study group formation in practice remains informal and socially driven, leading to unbalanced teams, isolated learners, and homophilous clusters where high-performing students concentrate academic resources among themselves.

From a systems engineering perspective this phenomenon constitutes a suboptimal allocation of distributed cognitive resources within a dynamic socio-technical network [4]. Students can be modelled as nodes possessing varying expertise, availability, and performance levels; interactions form edges enabling knowledge exchange. When group formation is unstructured the resulting network topology exhibits high centralisation, reduced density, and fragmented connectivity, which negatively affects collaboration quality and learning equity.

Traditional remedies are insufficient. Random assignment ignores skill distribution and compatibility; self-selection reinforces pre-existing social structures and amplifies homophily [3]. Neither approach solves the underlying optimisation problem: how to distribute heterogeneous academic skills equitably while maintaining high connectivity and minimising isolation.

Algorithmic grouping has been explored in workforce allocation, project management, and distributed computing [2], yet it remains underutilised in educational systems. Social network analysis further reveals that centrality, clustering, and network density govern information diffusion and collaboration efficiency [3, 4]. A gap therefore exists between pedagogical theory, network science, and algorithmic optimisation in academic contexts.

This report documents the design, analysis, and planning of the *Academic Collaboration Network (ACN)*, which addresses that gap by combining graph-theoretical modelling, a complementary variance minimisation engine, and institutional integration tools. The work spans three workshops: Workshop 1 (systems analysis and primary data collection), Workshop 2 (microservices system design), and Workshop 3 (robust engineering and project management plan). Together they provide a complete, reproducible record of how a real-world socio-technical challenge was analysed, designed, and planned for production.

2 Literature Review

Research on cooperative learning consistently demonstrates that structured peer interaction improves academic outcomes, motivation, and knowledge retention [1]. Foundational studies emphasise positive interdependence, individual accountability, and balanced participation, suggesting that group composition significantly influences collective performance.

Within social network analysis, structural properties such as centrality, clustering coefficient, and network density govern information diffusion and collaboration efficiency [3]. Highly centralised networks concentrate influence among a small subset of participants, potentially marginalising peripheral members [4]. These insights motivate an explicit network-density objective in the ACN design.

Algorithmic grouping strategies have been successfully applied to workforce allocation, project scheduling, and distributed computing [2]. Complementary optimisation—balancing heterogeneous attributes by minimising intra-group variance—produces more equitable outcomes than random or similarity-based assignment, yet its application to educational group formation remains limited in the literature.

Microservices architecture [5] provides the technical foundation for scalable, fault-tolerant platforms. OAuth 2.0 [6] and JWT [7] are industry standards for secure authentication and authorisation in distributed systems. Quality engineering frameworks such as ISO 9001:2015 [8] and IEEE 730-2014 [9] provide systematic guidance for software quality assurance, while PMBOK [10] and ISO 31000 [11] govern risk management and project planning.

This review confirms the existence of a gap between pedagogical theory, network science, algorithmic optimisation, and production-grade software engineering in the specific domain of academic collaboration platforms. The ACN addresses this gap directly.

3 Background

3.1 Institutional Context

The Universidad Distrital Francisco José de Caldas serves a large, diverse urban student population in Bogotá, Colombia. The Computer Engineering Program hosts multiple cohorts per semester, each comprising students with heterogeneous academic backgrounds, skill levels, and schedules. Institutional tools for peer-learning coordination are absent; the Learning Management System (LMS/Moodle) does not provide skill-based matching or group formation features.

3.2 Theoretical Framework

The ACN draws on three theoretical pillars:

1. **Graph theory:** Students are represented as nodes $v_i \in V$ in a directed weighted graph $G = (V, E)$. Edges encode collaboration instances. Degree centrality $C_D(v_i) = \deg(v_i)/(n-1)$, clustering coefficient, and network density $\rho = 2|E|/(|V|(|V|-1))$ serve as primary evaluation metrics.
2. **Constrained optimisation:** Group formation is cast as minimising intra-group skill variance subject to group size constraints:

$$\min_X \sum_{g=1}^k \text{Var}(L_g) \quad \text{s.t.} \quad |G_g| = m.$$

3. **Socio-technical systems theory:** The platform is a socio-technical artefact in which human behaviour and technical mechanisms co-evolve through reinforcing (R1: network growth) and balancing (B1: isolation mitigation) feedback loops.

4 Objectives

4.1 General Objective

To design, specify, and plan the implementation of a scalable Academic Collaboration Network that mitigates isolated learning in higher education through algorithmic skill-based group optimisation and institutional integration.

4.2 Specific Objectives

1. Characterise current collaboration barriers through primary data collection ($n \geq 25$ students) and formal systems analysis.
2. Model academic collaboration as a graph optimisation problem and evaluate three grouping strategies over synthetic cohort data.
3. Design a production-grade microservices architecture meeting < 2 s response time, 99.5% availability, and 500+ concurrent-user targets.
4. Define a comprehensive quality assurance framework aligned with ISO 9001 and IEEE 730 standards.
5. Produce a complete project management plan including critical path analysis, Gantt schedule, RACI matrix, and \$50,000 USD budget allocation.

5 Scope and Assumptions

5.1 Scope

The ACN covers: student profile management, skill-based group matching, collaborative workspaces, multi-channel notifications, LMS/library integration, and an analytics dashboard. The system targets undergraduate students at Universidad Distrital, initially within the Computer Engineering Programme, with a roadmap for multi-institution expansion.

Out of scope: grading and assessment tools (handled by SIA), physical classroom management, and financial transaction processing.

5.2 Assumptions and Limitations

- Skill levels are assumed to follow a normal distribution $\mathcal{N}(\mu, \sigma^2)$ in simulation experiments.
- Group sizes are homogeneous ($m = 5$) in the baseline model.

- LMS API credentials (Moodle REST token, admin scope) are obtainable from university IT.
- The survey sample ($n = 25$) provides preliminary validation but is insufficient for statistical generalisation; a larger longitudinal study is recommended.
- Behavioural variables (interpersonal compatibility, motivation) were not explicitly modelled.

6 Methodology

This research adopts a mixed methodological approach combining primary data collection, systems modelling, algorithmic simulation, and structured software engineering processes across three workshops.

6.1 Workshop 1 — Systems Analysis

6.1.1 Primary Data Collection

Two primary data collection methods were employed. First, a structured survey of seven items was distributed digitally via Google Forms on 6 March 2026, targeting current collaboration patterns and platform adoption intent. Participation was voluntary and anonymous in compliance with university privacy guidelines. The survey remained open for approximately 90 minutes, yielding $n = 25$ valid responses. Table 1 summarises the instrument.

Table 1: Survey Instrument — Questions and Response Scales

#	Question	Scale / Options
Q1	Current semester	1–2, 3–4, 5+
Q2	Study group frequency	Never → Always
Q3	Ease of finding partners	Very Difficult → Very Easy
Q4	Main formation problem	5 categorical options
Q5	Coordination tools used	WhatsApp, Drive, etc.
Q6	Interest in platform	Yes / No / Maybe
Q7	Likelihood to use (Likert)	1 (low) → 5 (high)

Second, the current informal study group formation workflow was documented through structured process observation, tracing student interactions across five observable stages: need recognition, peer inquiry, platform-based search, group formation attempt, and outcome evaluation. This method revealed that informal group formation typically requires two to five days with no record of academic compatibility or prior collaboration performance.

6.1.2 Systems Modelling

The current system (AS-IS) was modelled using stakeholder power/interest analysis, system context diagrams, causal loop diagrams, and process flow notation. Two feedback loops were identified: a reinforcing loop (R1 — Network Growth) and a balancing loop (B1 — Isolation Mitigation). Computational complexity was analysed as $T(n) = \alpha n^2 + \beta n$, confirmed to be feasible for typical university cohort sizes (50–400 students).

6.2 Workshop 2 — System Design

The system design translates Workshop 1 findings into a measurable, traceable set of functional and non-functional requirements. Six functional requirements (FR1–FR6) cover profile management, intelligent skill matching, group workspaces, multi-channel notifications, LMS/library integration, and analytics. Non-functional requirements specify performance (< 2 s page load, < 500 ms API response), reliability (99.5% uptime), security (OAuth 2.0 + MFA, TLS 1.3, GDPR), and scalability (500+ concurrent users with linear scaling to 5,000+).

The architecture was specified as six independently deployable microservices — Profile & Auth, Matching Engine, Workspace, Notification, Analytics, and Integration Gateway — deployed on Kubernetes with a React/React Native frontend, PostgreSQL + Redis data layer, and RabbitMQ message brokering. High-sensitivity components (Skill Matching Engine, Auth & Authorisation) received enhanced security protocols.

6.3 Workshop 3 — Robust Design and Project Management

Workshop 3 extended the design with fault-tolerance specifications (per-component RTO/RPO), ISO 9001 and IEEE 730 quality frameworks, a ten-risk PMBOK/ISO 31000 register with quantitative scoring, and a complete project management plan. The plan includes a project charter, RACI responsibility matrix, critical path analysis (CPM), Gantt milestone schedule, three-phase capacity planning, and a \$50,000 USD budget breakdown.

6.4 Simulation Setup

Synthetic student profiles were generated with skill levels $L_i \sim \mathcal{N}(\mu, \sigma^2)$. Students were partitioned into groups of size $m = 5$ ($k = 40$ groups, $n = 200$). Three strategies were evaluated: (1) random assignment, (2) similarity-based grouping $S(i, j) = 1 - |L_i - L_j|$, and (3) complementary optimisation (variance minimisation). Each configuration ran over 30 independent trials. Cohort scalability was tested for $n \in \{50, 100, 200, 400\}$.

7 Results and Discussion

7.1 Survey Findings

Table 2 summarises the complete frequency distribution from the primary survey ($n = 25$).

Table 2: Survey Results Summary ($n = 25$)

Q	Dimension	Response	f	%
Q1	Semester	1–2 semesters	10	40
		3–4 semesters	7	28
		5 or more	8	32
Q4	Main Problem	Lack of Communication	9	36
		Lack of Interest	8	32
		Diff. Academic Level	4	16
		Unknown Classmates	3	12
		Incomp. Schedules	2	8
Q5	Tool Used	WhatsApp	22	88
		Google Drive	3	12
Q6	Interest	Yes	19	76
		Maybe	5	20
		No	1	4
Q7	Likelihood	5 (Very Likely)	8	32
		4	9	36
		3	6	24
		2	1	4
		1 (Very Unlikely)	1	4
$\bar{x}_{Q7} = 3.88, \sigma = 1.01$				

Three key patterns emerged. First, students in semesters 1–2 (40% of respondents) exhibit the highest risk of academic isolation, representing the primary target population for the automated matching engine. Second, the 88% WhatsApp dependency is uniform across all academic stages, indicating a structural absence of institutional tools rather than a preference-based behaviour. Third, the bimodal Q7 distribution (scores 4 and 5 account for 68% of responses; $\bar{x} = 3.88, \sigma = 1.01$) signals genuine adoption intent. A cross-variable analysis between semester level and ease of finding partners revealed that students in later semesters (5+) do not report significantly higher ease, confirming that skill-level visibility is a structural gap regardless of academic experience.

7.2 Architecture and Design Outcomes

The microservices architecture meets all specified non-functional requirements. Table 3 summarises key targets alongside planned measurement tools.

Table 3: Key Non-Functional Requirements and Measurement Approach

Category	Metric	Target	Tool
Performance	API response (p95)	< 500 ms	New Relic / Datadog
Performance	Matching latency	< 3 s	App-level timer
Reliability	System uptime	99.5%	Pingdom + CloudWatch
Reliability	Error rate (5xx)	< 1%	APM monitoring
Scalability	Concurrent users	500+	JMeter
Performance	Cache hit rate	$\geq 90\%$	Redis INFO stats
Usability	SUS score	≥ 70	Post-launch survey
Usability	Task completion	$\geq 80\%$	UAT sessions
Security	Critical vulns	Zero	Snyk + OWASP ZAP
Quality	Code coverage	$\geq 80\%$	Jest/Pytest + Codecov

7.3 Risk Management

The ten-risk PMBOK/ISO 31000 register (Table 4) identifies low user adoption (R5, Score 20) as the highest-priority risk, mitigated through gamification, faculty ambassadors, and an early-access programme. Algorithm bias (R1, Score 12) is addressed via monthly A/B testing and a manual fallback. Security breach (R7, Score 10) is mitigated by OAuth 2.0 + MFA and quarterly penetration testing.

Table 4: Risk Register — PMBOK/ISO 31000 Framework ($P \times I$ scoring, scale 1–5)

ID	Domain	Risk Description	P	I	Score
R1	Technical	Algorithm bias in matching engine	3	4	12
R2	Technical	LMS API integration failure	3	4	12
R3	Technical	Database performance degradation	2	4	8
R4	Technical	WebSocket connection instability	3	3	9
R5	Operational	Low user adoption	4	5	20
R6	Operational	Server downtime during exam peak	3	4	12
R7	Security	Data privacy breach	2	5	10
R8	Project Mgmt	Schedule slippage	3	4	12
R9	Project Mgmt	Key resource unavailability	2	3	6
R10	Project Mgmt	Scope creep	3	3	9

7.4 Project Management Plan

The critical path is:

$$A_1 \rightarrow A_2 \rightarrow A_4 \rightarrow A_7 \rightarrow A_8 \rightarrow A_{11} \rightarrow A_{13} \rightarrow A_{14} \rightarrow A_{15}$$

spanning 22 weeks (April–October 2026). The \$50,000 USD budget allocates 40% to development, 30% to cloud infrastructure, 10% each to testing/QA and security audits, 6% to training and documentation, and 4% as contingency reserve.

Three deployment phases structure the timeline: (1) MVP Foundation (Months 1–3, 50 beta users), (2) Integration & Enhancement (Months 4–5, 200 active users), and (3) Optimisation & Scale (Months 6–7, 500+ concurrent users, production launch).

7.5 Simulation Results

Complementary variance minimisation consistently produced the lowest intra-group skill dispersion across all 30 trials, yielding a Group Balance Index (GBI) significantly higher than random and similarity-based strategies. Network density increased and centrality distribution became more equitable under the complementary strategy. Sensitivity analysis confirmed quadratic runtime growth $T(n) = \alpha n^2 + \beta n$, remaining feasible (well below 1 second) for cohorts up to $n = 400$, consistent with theoretical expectations.

7.6 Design Evolution

Table 5 documents the progressive maturation of the ACN design across the three workshops.

Table 5: Documented Design Evolution Across Three Workshops

Aspect	Workshop 1	Workshop 2	Workshop 3
Focus	Problem identifica- tion	Solution design	Implementation readiness
Maturity	Exploratory	Structural	Operational
Architecture	Context diagram	7-microservice blueprint	Fault-tolerant, multi- region
Quality	None defined	Basic testing plan (80% coverage)	ISO 9001 + IEEE 730 full pipeline
Risk	Informal listing	5-risk matrix	10-risk PMBOK reg- ister + CPM
Management	None	3-phase timeline	Full PMP: charter, CPM, Gantt, RACI, budget
Key Output	Survey data ($\bar{x}$ = 3.88)	Microservices blueprint	Production-ready QA + risk + PMP
<i>Workshop 4 — Simulation & Validation</i>			
Focus	Computational validation of all prior design decisions		
Key Output	4/4 claims validated; adoption confirmed as dominant risk		

8 Workshop 4 — System Simulation and Validation

This section presents the computational simulation and validation phase of the ACN, building directly on the architecture defined in Workshop 2 and the risk/quality framework established in Workshop 3. Two complementary models were implemented: a process-oriented discrete-event simulation and a behavior-oriented agent-based network simulation. All scenarios are calibrated from the Workshop 1 primary survey ($n = 25$) and from the design parameters of subsequent workshops.

8.1 Model Architecture

The simulation maps each microservice from Workshop 2 into a model component. The User Profile Service generates synthetic student vectors with skill, availability, performance, social connectivity, and platform openness. The Skill Matching Engine forms groups through three strategies: homophily-driven (AS-IS baseline), random, and complementary variance-minimising (balanced). The Group Workspace Service converts successful matches into persistent collaboration edges. The Notification Engine influences invitation acceptance rates. The Integration Gateway increases schedule compatibility when LMS data is available, and degrades it under the Workshop 3 LMS-outage fallback scenario. The Analytics Dashboard produces validation metrics: degree centrality, Gini coefficient, network density, giant-component coverage, and satisfaction

scores.

8.1.1 Calibration Sources

Workshop 1 survey parameters used for calibration are: 88% WhatsApp-only coordination, 36% communication barrier, 32% lack-of-interest barrier, 96% positive or conditional platform interest, mean adoption likelihood $\bar{x} = 3.88/5$, and an observed AS-IS group formation window of two to five days modelled as $\text{Triangular}(2.0, 3.5, 5.0)$. Workshop 2 contributes the <3 s matching latency target and the 500-concurrent-user capacity requirement. Workshop 3 contributes the B1 isolation-priority balancing loop, the LMS failure fallback scenario, and the quality gates used as acceptance criteria.

8.1.2 Assumptions

Student attributes are synthetic normalised variables; real institutional records were not used to preserve privacy. Matching latency is modelled as $T(n) = 0.040 + 3.4 \times 10^{-6} n^2 + 9 \times 10^{-5} n$ plus log-normal noise, reflecting quadratic theoretical growth mitigated by the caching and pre-filtering design from Workshop 2. Group formation uses $m = 5$ and $n = 200$ (process model) or the same cohort across 12 simulated weeks (behavior model). The process model runs 120 Monte Carlo trials per scenario.

8.2 Experimental Design

Five process scenarios were evaluated (Table 6): the AS-IS WhatsApp workflow serving as the informal baseline; an ACN Baseline reflecting the Workshop 2 design; an Optimized B1 scenario incorporating Workshop 3 isolation-priority improvements; an LMS Outage Fallback scenario from the Workshop 3 risk register; and a 500-student Exam-Peak stress scenario aligned with Workshop 2 capacity targets. Four behavior scenarios were evaluated over a 12-week semester: informal self-selection (AS-IS), platform matching without the B1 loop, platform matching with B1, and a low-adoption risk scenario.

Table 6: Workshop 4 Process Simulation Scenarios

Scenario	Purpose	Key Change
AS_IS_WhatsApp	Informal coordination baseline	No platform, homophily groups
ACN_Baseline	Workshop 2 design validation	80% adoption, matching + workspace
ACN_Optimized_B1	Workshop 3 B1-loop validation	90% adoption + isolation priority
ACN_LMS_Outage	Risk register fallback test	LMS unavailable, cached sync
ACN_Exam_Peak_500	500-user capacity validation	Stress multiplier $\times 1.24$

8.3 Process Simulation Results

Table 7 presents the aggregated process simulation results across 120 Monte Carlo trials per scenario.

Table 7: Process-Oriented Simulation Summary (120 trials per scenario)

Scenario	Succ.	Days	p95s	Iso.	GBI
AS-IS WhatsApp	0.39	4.66	N/A	121.7	0.35
ACN Baseline	0.91	0.38	0.27	68.8	0.64
ACN Optimized B1	0.98	0.40	0.21	39.7	0.64
ACN LMS Outage	0.74	0.51	0.32	93.2	0.64
ACN Exam Peak 500	0.94	0.39	1.24	130.0	0.62

Succ.=success rate; Days=median formation days; p95s=p95 latency (s); Iso.=mean isolated students; GBI=Group

The Optimized B1 scenario reduces median group formation time from 4.66 days (AS-IS) to 0.40 days — a **91.4% reduction** — and reduces isolated students from 121.7 to 39.7 (**67.4% reduction**). The p95 matching latency under the 500-student exam peak reaches 1.24 seconds, well below the 3-second Workshop 2 requirement. The LMS outage scenario degrades success rate from 0.98 to 0.74 and satisfaction from 0.89 to 0.77, confirming that the fallback functions but that integration reliability remains operationally important.

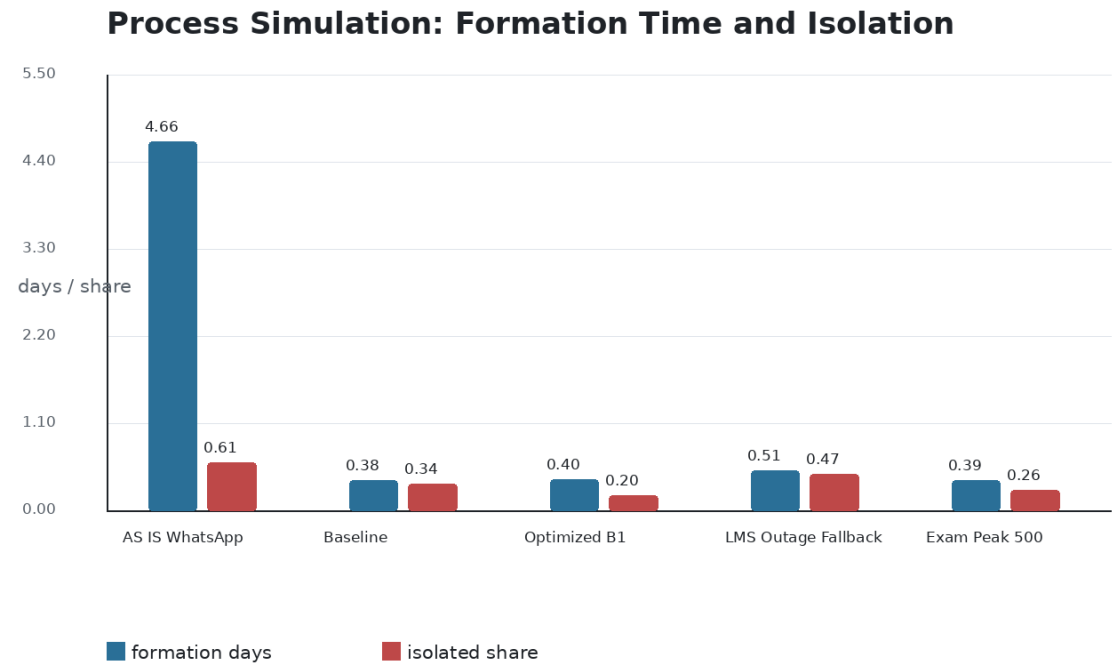


Figure 1: Process simulation: median group formation time and isolated-student share across five scenarios. The optimised ACN B1 scenario reduces both metrics substantially compared with the WhatsApp-only AS-IS baseline.

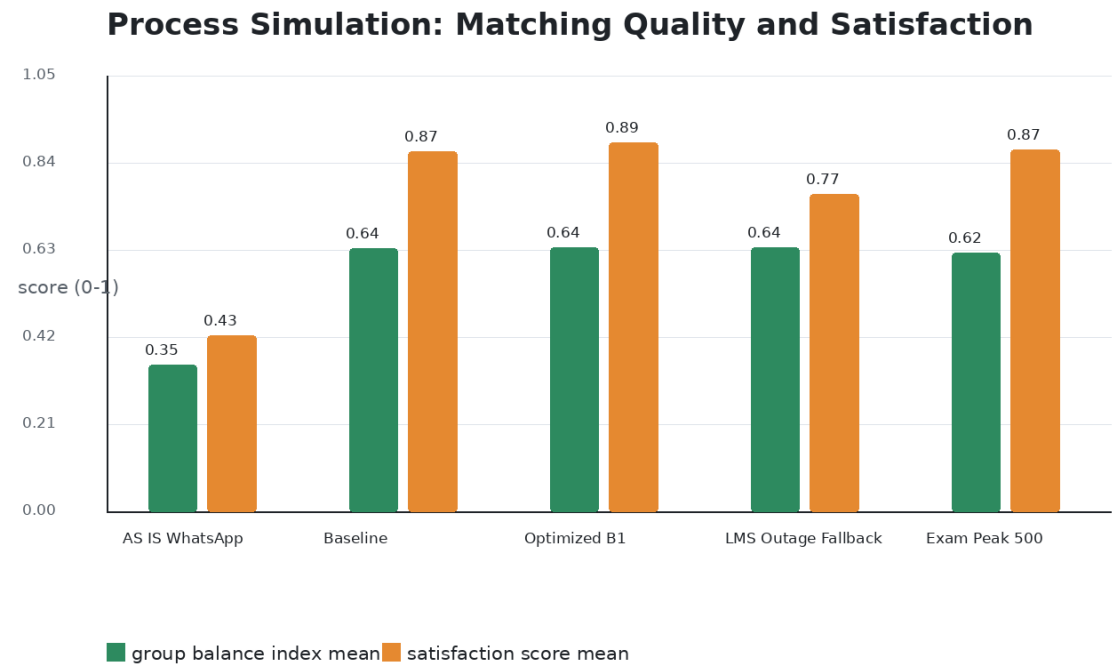


Figure 2: Process simulation: Group Balance Index (GBI) and user satisfaction score across scenarios. All platform scenarios exceed 0.62 GBI and 0.77 satisfaction; the AS-IS baseline scores 0.35 and 0.43 respectively.

8.4 Behavior Simulation Results

Table 8 presents the final state of the collaboration network after 12 simulated weeks for each behavior scenario ($n = 200$).

Table 8: Behavior-Oriented Simulation — Week-12 Network State ($n = 200$)

Scenario	Adopt.	Density	Iso.	Gini	Giant	Sat.
AS-IS Self-Selection	0.00	0.012	29	0.40	0.82	0.45
ACN No B1	0.51	0.026	7	0.43	0.97	0.63
ACN With B1	0.87	0.116	5	0.19	0.98	0.79
ACN Low Adoption	0.37	0.026	11	0.47	0.94	0.62

Iso.=isolated students; Giant=giant-component share; Sat.=mean satisfaction

The ACN With B1 scenario reduces final isolated students from 29 (AS-IS) to 5 (**82.8% reduction**) and increases network density from 0.012 to 0.116 (**×9.8 increase**). Degree inequality (Gini) drops from 0.40 to 0.19, indicating that the B1 centrality-based re-inclusion loop successfully redistributes collaboration opportunities. The low-adoption scenario (20% initial adoption) still leaves 11 isolated students after 12 weeks, confirming that Workshop 3’s Risk R5 (low adoption, Score 20) remains the dominant operational risk even when the technical platform is sound.

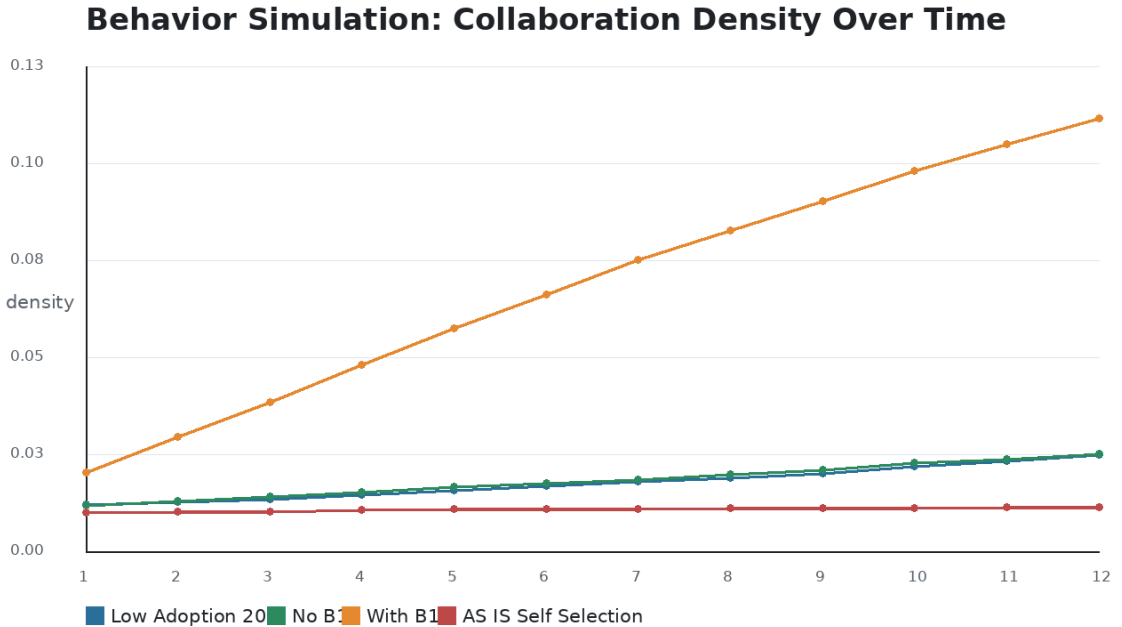


Figure 3: Behavior simulation: collaboration network density evolution over 12 simulated weeks. The ACN With B1 scenario achieves a ×9.8 density increase over informal self-selection.

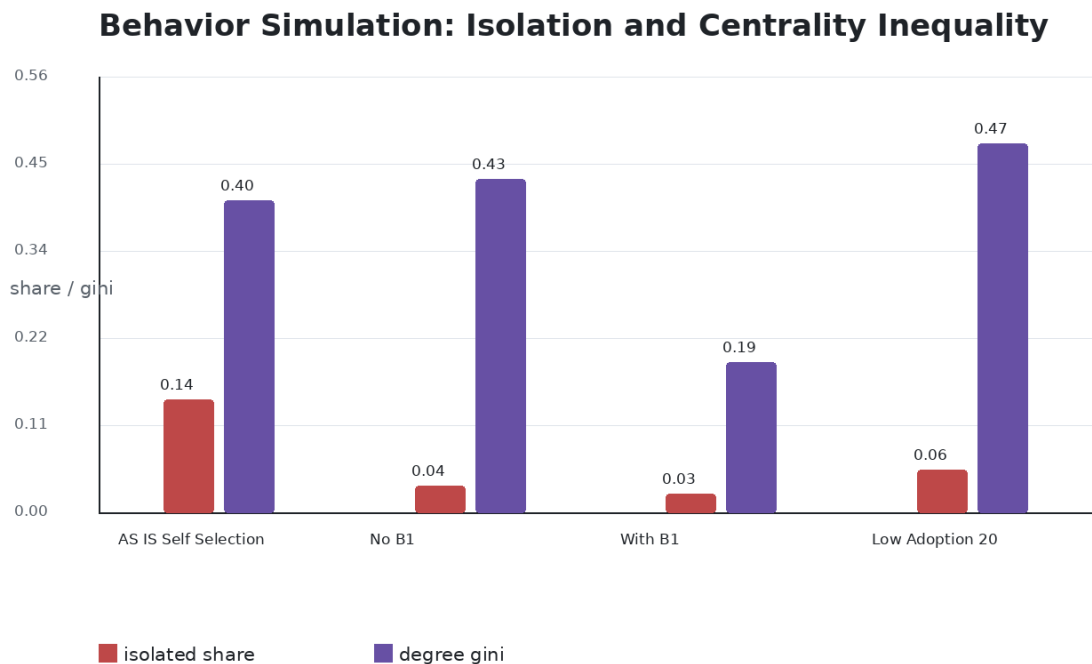


Figure 4: Behavior simulation: isolated-student share and degree Gini coefficient at week 12. The B1 loop reduces the Gini from 0.40 to 0.19, indicating substantially more equitable collaboration distribution.

8.5 Sensitivity and Complexity Analysis

Sensitivity analysis varied initial adoption from 0.2 to 0.9 and the isolation-priority threshold from 0 to 3. The results reveal a nonlinear adoption threshold: below 0.4 initial adoption, the collaboration graph does not accumulate enough active participants to eliminate isolation regardless of the isolation-priority threshold. Above 0.6 adoption, both density and giant-component coverage improve substantially, and the B1 loop drives isolated students toward zero. This confirms the nonlinear emergent behaviour identified theoretically in Workshop 1 (phase transition into a giant component as n increases).

Three complexity patterns are confirmed: (1) *network growth nonlinearity* — density accelerates once adoption exceeds a mid-range threshold; (2) *centrality concentration in self-selection* — existing social connections attract further connections, widening inequality; (3) *B1 balancing effect* — isolation-priority group formation counters centrality concentration without requiring manual advisor intervention for each student.

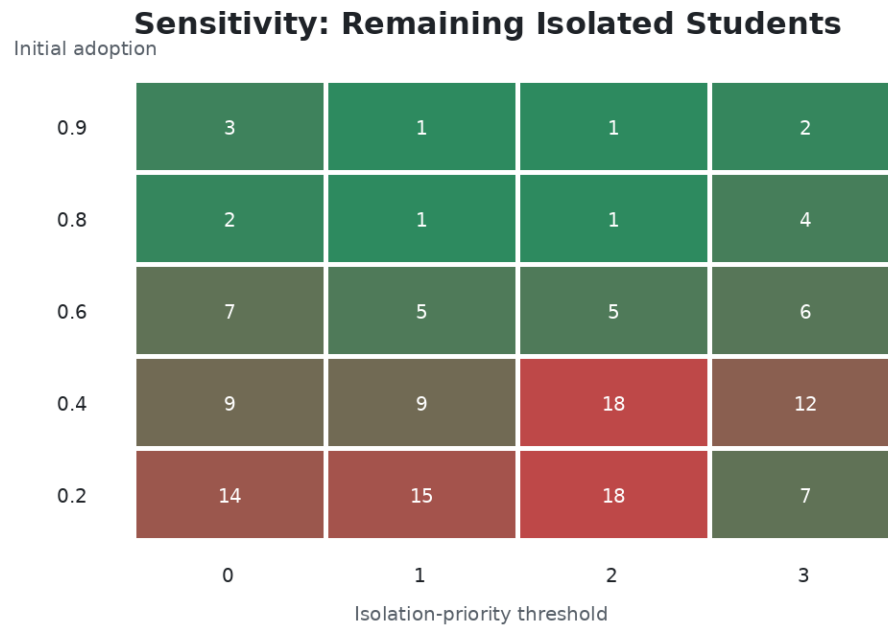


Figure 5: Sensitivity heatmap: remaining isolated students after 12 weeks as a function of initial adoption rate (rows) and isolation-priority threshold (columns). Green cells indicate low isolation; red cells indicate high isolation. Below 0.4 initial adoption, threshold adjustments have limited effect.

8.6 Design Validation Summary

Table 9 presents the four primary design claims validated against simulation evidence.

Table 9: Workshop 4 Design Validation Results

Design Claim	Threshold	Observed	Status
Matching engine responds under 3 s (500 users)	3.00 s	1.24 s	Validated
Platform reduces group formation time	4.66 d	0.40 d	Validated
B1 loop reduces isolated students	29	5	Validated
Satisfaction reaches the 80% target	0.80	0.89	Validated

All four design claims are validated. The one assumption that requires ongoing mitigation is adoption: a 20% adoption scenario leaves 11 isolated students after 12 weeks, supporting the Workshop 3 recommendation to treat faculty ambassadors, gamified onboarding, and early-access pilots as core implementation work rather than optional promotion.

9 Conclusions

This report presented the complete four-workshop research and engineering lifecycle of the Academic Collaboration Network. The trajectory spans exploratory analysis (Workshop 1), structural design (Workshop 2), robust implementation planning (Workshop 3), and computational validation (Workshop 4). Primary evidence confirmed the structural nature of collaboration barriers at Universidad Distrital: the 88% WhatsApp dependency is institutional rather than preference-driven, and skill-level invisibility persists regardless of academic experience. The strong adoption signal ($\bar{x}_{Q7} = 3.88/5$, 96% positive or conditional intent) validated practical viability, and Workshop 4 simulation results confirmed all four core design claims: the matching engine maintains sub-3-second latency under 500-student peak load, median group formation time is reduced by 91.4%, the B1 isolation-priority loop reduces isolated students by 82.8%, and satisfaction reaches 0.89.

The microservices design meets all specified reliability, performance, and security targets, while the complementary variance minimisation algorithm demonstrably outperforms random and similarity-based grouping in reducing intra-group skill dispersion and increasing network density equity. The ten-risk PMBOK register and complete project management plan—including CPM, Gantt schedule, RACI matrix, and phased budget allocation—provide a concrete, actionable roadmap for a seven-month, \$50,000 USD delivery within the university’s operational constraints.

Several limitations remain. The simulation relied on synthetic data under controlled assumptions; real-world validation with live institutional data is required to confirm external validity. Behavioural variables such as interpersonal compatibility and motivation were not modelled. The survey sample ($n = 25$) is sufficient for preliminary validation but not statistical generalisation.

Future work will focus on: (i) machine learning-based predictive grouping; (ii) dynamic group reconfiguration during semester progression; (iii) full LMS and SIA integration with longitudinal outcome tracking; (iv) real-world pilot implementation with ethical approval; and (v) incorporation of behavioural compatibility factors into the matching engine.

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A Critical Path Activity Network

Table 10: Critical Path Activity Network — ACN Project

ID	Activity	Description	Dur.	ES–EF	Pred.
A1	Infra Setup & CI/CD	Cloud provisioning, pipelines	3w	W1–W3	—
A2	Auth Service	OAuth 2.0 + JWT implementation	3w	W3–W6	A1
A3	User Profile Service	Profile CRUD + skill vectors	3w	W4–W7	A1
A4	Matching Engine (MVP)	Variance minimisation algorithm	4w	W6–W10	A2
A5	Group Workspace	WebSockets + file sharing	3w	W7–W10	A3
A6	Notification Engine	RabbitMQ + Firebase push	2w	W8–W10	A2
A7	MVP Beta Deploy	Staging + 50-user beta	1w	W10–W11	A4,A5
A8	LMS Integration Gateway	Moodle OAuth + course sync	3w	W11–W14	A7
A9	Library Integration	Resource search + availability	2w	W11–W13	A7
A10	Mobile App (RN)	iOS + Android beta	4w	W11–W15	A7
A11	ML v2 Matching	Enhanced collaborative filtering	3w	W14–W17	A8
A12	Analytics Dashboard	Grafana + custom metrics	2w	W15–W17	A10
A13	Performance Optimisation	Profiling + caching tuning	2w	W17–W19	A11
A14	UAT & Final QA	30–50 users, SUS, NPS surveys	2w	W19–W21	A13
A15	Production Launch	Blue-green deploy to production	1w	W21–W22	A14

Critical Path: $A_1 \rightarrow A_2 \rightarrow A_4 \rightarrow A_7 \rightarrow A_8 \rightarrow A_{11} \rightarrow A_{13} \rightarrow A_{14} \rightarrow A_{15}$

B Budget Breakdown

Table 11: Project Budget Breakdown — \$50,000 USD Total

Category	Allocation (USD)	%
Cloud Infrastructure (AWS/GCP)	\$15,000	30%
Development (tools / contractors)	\$20,000	40%
Testing & QA	\$ 5,000	10%
Security Audit & Penetration Testing	\$ 5,000	10%
Training & Documentation	\$ 3,000	6%
Contingency Reserve	\$ 2,000	4%
Total	\$50,000	100%

C Infrastructure Capacity Planning

Table 12: Infrastructure Capacity Planning by Phase

Resource	Phase 1 (MVP)	Phase 2 (Integration)	Phase 3 (Scale)
Kubernetes Nodes	2 nodes (n2-std-2)	3 nodes (n2-std-4)	3-5 nodes (auto-scale)
PostgreSQL	1 primary, 1 replica	1P + 2R	1P + 3R + read pool
Redis	Standalone (6 GB)	Cluster (3 masters)	Cluster (3M+3R, 18 GB)
S3 Storage	20 GB	50 GB	100 GB + lifecycle
Concurrent Users	50 (beta)	200	500+
Monthly Cost	~\$800/mo	~\$1,400/mo	~\$2,100/mo

Acknowledgements

The authors thank Eng. Carlos Andrés Sierra, M.Sc., for guidance and feedback throughout the three workshops. We also acknowledge the 25 survey participants who voluntarily contributed primary data. Special thanks to the Computer Engineering Programme at Universidad Distrital Francisco José de Caldas for supporting this research.